

QIU-HAN GU

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Research Interests

Primary Interests: Formal methods for trustworthy AI systems, with a focus on the principled verification of critical properties such as safety and robustness. I also work on machine learning for automated program synthesis, a complementary direction that supports scalable formal reasoning.

Other Interests: Neural Network Verification, Probabilistic Programming Verification, Constrained Generation Optimization.

Education

University of Illinois Urbana-Champaign

Master Science of Computer Science

Advisor: Prof. Gagandeep Singh

Aug.2024 – May.2026(Expected)

Champaign, United States

Nanjing University

Bachelor of Science in Computer Science and Technology

Advisor: Prof. Linzhang Wang and Prof. Yu Wang

Sept.2020 – Jun.2024

Nanjing, China

Preprints and Publications

Qiuhan Gu, Avaljot Singh, Gagandeep Singh, “SAIL: Sound Abstract Interpreters with LLMs” PLDI, 2026. [\[paper\]](#) [\[code\]](#)

Qiuhan Gu, “LLM-Based Code Generation Method for Golang Compiler Testing,” SRC@FSE (1st place), 2023. [\[paper\]](#) [\[code\]](#)

Research Experience

Cost-Driven Synthesis of Sound Abstract Interpreters

Research Assistant. Advisor: Prof. Gagandeep Singh

Jun.2025 – Nov.2025

FOCAL Lab, University of Illinois

- Proposed the first unified framework to automatically synthesize globally sound abstract interpreters across multiple abstract domains.
- Formalized the synthesis process as a constrained optimization problem over an infinite search space of abstract transformers and introduced a novel mathematically grounded cost function that quantitatively measures unsoundness deviation to guide refinement.
- Empirically demonstrated the framework’s capability to match handcrafted transformer quality and, more importantly, synthesize globally sound and precise transformers for complex nonlinear operators absent from the literature.
- Led to a first-author paper submitted to PLDI 2025, currently under review.

Enhancing DL System Fuzzing via LLMs

Intern. Advisor: Prof. Lingming Zhang and Prof. Darko Marinov

Sept.2024 – Dec.2024

University of Illinois

- Developed an automated pipeline to generate and validate deep learning operator constraints using large language models, improving operator validity from 46.3% to 57.6%.
- Designed a domain-specific language to represent operator constraints and incorporated symbolic verification techniques to ensure correctness.
- Integrated LLM-generated operators into NNSmith and NeuRI fuzzing frameworks, achieving higher coverage and discovering multiple compiler bugs.

LLM-Based Code Generation Method for Golang Compiler Testing

Undergraduate Researcher. Advisor: Prof. Yu Wang

Sept.2023 – Dec.2023

SEG, Nanjing University

- Implemented a novel LLM-based high-quality code generation method for the Golang compiler, generating testcases with 3.38% average coverage and only 2.79% of them had syntax errors.

- Published a paper as the first author at ESEC/FSE Conference 2023, LLM-Based Code Generation Method for Golang Compiler Testing.
- Won 1st prize in the undergraduate division of ESEC/FSE'23 Student Research Competition.
- Kept on exploring the software testing technique and improving the performance of program analysis by machine learning.

LLM-Based Code Classification Method (Bachelor Thesis)

Feb.2024 – Jun.2024

Undergraduate Researcher. Advisor: Prof. Yu Wang

SEG, Nanjing University

- Implemented an LLM-based high-quality code classification method of Go language, achieving an accuracy of 96.4%.
- Combine large language models and traditional methods to efficiently model code in terms of lexical, syntactic, and semantic aspects.
- Defined custom classification categories, providing fine datasets for downstream tasks.

Automatic Detection of Intracranial Aneurysms Based on Deep Learning

May.2022 – Apr.2023

Undergraduate Researcher.

Nanjing University

- Built a clinically applicable deep learning system for automated intracranial aneurysm detection from CTA images.
- Implemented and deployed a unified web platform allowing medical institutions to access and run the diagnostic system.
- Created a comprehensive implementation plan covering system design, deployment pipeline, and business integration.
- Won the first prize in the national competition (1/154)

Honors & Awards

- **ESEC/FSE'23 Student Research Competition, 1st Prize (Undergraduate Division).** San Francisco, United States. Dec.2023.
- **Huawei Scholarship.** Nanjing University, China. Oct.2023.
- **National Scholarship of Undergraduate.** China. Dec.2021.
- **National College Student Entrepreneurship Competition, 1st Prize.** China. Mar.2023.

Teaching Experience

- University of Illinois Urbana-Champaign: CS427 Software Engineering, 2025 Fall (*Teaching Assistant*)
- University of Illinois Urbana-Champaign: CS101 Introduction to Computer Science, 2025 Spring (*Teaching Assistant*)
- University of Illinois Urbana-Champaign: CS427 Software Engineering, 2024 Fall (*Teaching Assistant*)
- Nanjing University: Advanced Python Programming, 2024 Spring (*Teaching Assistant*)
- Nanjing University: Advanced Python Programming, 2023 Fall (*Teaching Assistant*)

Other Skills

Technologies: Git, Docker, Anaconda, SQL, vLLM, Z3 SMT Solver, ANTLR4, DSL Development, Web Development, App Development, Machine Learning, Formal Methods

Programming Languages: Python, C/C++

Libraries: PyTorch, TensorFlow, HuggingFace Transformers, NumPy, Pandas, Matplotlib

Course Work: Formal Methods, Machine Learning, Security and Privacy, Operating System, Software Engineering, Natural Language Processing, Computer Graphics, Computer Vision, Data Structure and Algorithms, Linear Algebra, Calculus